## Verification of serial outputs from the STM32Discovery Board implementing pqm4 library and ML-KEM-768 algorithm, with official KAT.

On the STM32F4 Discovery / libopencm3 platform, PQM4's default randombytes_init() is intended to be non-deterministic and sourced from the board's hardware RNG (TRNG), and hence the output changes every run. Therefore, the output we got from the implementation will not match the KAT.
To tackle this, we use a DRBG (Deterministic Random Bit Generator) because NIST KAT validation requires the exact same "randomness" stream every time., Thus needed to mplement the same deterministic DRBG used in the KAT generator (common is AES-CTR-DRBG with derivation function, like in NIST PQC KAT code)

### 1) Added deterministic NIST AES-CTR-DRBG (shim for KAT/randombytes_init function)

Create New file: **common/kat_rng_shim.c**
What it does:

- Implements randombytes_init() using the NIST AES-CTR-DRBG (seeded with 48 bytes) instead of the default 32-byte hardware RNG.
- Provides a weak PQCLEAN_randombytes(), so NO_RANDOMBYTES builds still link, but the real RNG overrides when present.

### 2) Link the shim into STM32 builds

Modified: **mk/opencm3.mk**
Change: Added common/kat_rng_shim.c to LIBHAL_SRC so it links into STM32 builds.

```
# SPDX-License-Identifier: Apache-2.0 or CC0-1.0
LIBHAL_SRC := \
        common/hal-opencm3.c \
        common/randombytes.c \
        common/kat_rng_shim.c
```

### 3) Fix randombytes symbol resolution

Modified: **common/randombytes.c**
Change: Added a randombytes() function, and the wrapper.

randombytes() forwards to PQCLEAN_randombytes() (avoids macro recursion/link errors).

## 4) ACVP testvector flow for ML-KEM-768 (inputs only)

Modified: **mupq/crypto_kem/testvectors.c**

*The MUPQ framework provides common files and a hardware abstraction layer to ensure consistency across different microcontroller architectures*

Key updates:

- Added MLKEM_ACVP path (inside main) that uses inputs only (d, z, ek, m) from ACVP prompt.json.

```
#ifdef MLKEM_ACVP
 /* ACVP FIPS203 vectors: ML-KEM-768, tcId 26 */
#inputs for key generation
 const char *d_hex =
   "A2B4BCA315A6EA4600B4A3…";
 const char *z_hex =
   "D6BF055CB7B375E3271ED13…";

#inputs for encapsulation/decapsulation
 const char *ek_in_hex =
   "F255CE47334283B8622BE7CE…;
 const char *m_hex =
   "5BD922AF345AB90F297D0A82…";
```

- Computes and prints outputs (ek/dk, ct/ss) without hard-coded expected values.
- Uses indcpa_keypair_derand() and indcpa_enc() to match FIPS-203 derivation.
- Keeps original seed-based KAT flow under #else for non-ACVP builds.

## 5) Enable ACVP mode only for ML-KEM-768 m4fspeed testvectors

Modified: **mupq/mk/schemes.mk**

Change: Adds -DMLKEM_ACVP only for crypto_kem_ml-kem-768_m4fspeed_testvectors.elf

```
# This template defines all the targets for a scheme: a library file containing
# all the compiled objects, and an elf file for each test.
define schemelib
obj/lib$(2).a: $(call objs,$(call schemesrc,$(1)))
libs: obj/lib$(2).a
elf/$(2)_%.elf: CPPFLAGS+=-I$(1)
elf/crypto_kem_ml-kem-768_m4fspeed_testvectors.elf:
CPPFLAGS+=-DMLKEM_ACVP
elf/$(2)_%.elf: MUPQ_NAMESPACE=$(call namespace,$(2),$(3))
elf/$(2)_%.elf: PROFILE_HASHING=$$(filter %_hashing.elf,$$@)
elf/$(2)_%.elf: NO_RANDOMBYTES=$$(filter %_testvectors.elf,$$@)
```

Make sure to put correct values from **prompt.json**. Build and flash testvectors.elf to the board.

```
$ st-flash write bin/crypto_kem_ml-kem-768_m4fspeed_testvectors.bin
0x8000000
```

The serial output from the board then matches exactly with **expected.json** from KAT

The output looks like this for test input id **26**

```
HELLO FROM MODIFIED TESTVECTORS

keygen tcId = 26

ek =
5219C4CC17C35A828F3E21B2AB7496805C99EE041FCA0158A3314F07D053F364C887A6825958A62
5965D4885C2CB355E83A3C1BBB15446F891D2D24F145632CF06A5EE1A278CD3064A79AD5319385
3E4CEA654448A4297CEA3C9E87561629680F588953B858074292ED31C20DDD983E805D07BB9CFA
DD823C7900B604286C0184738CA04E0DA8289540E329605EFAA5960AAC0FD0760006C1F1993426
CC7BEA22BCBB3CC02E099B828E82F94045DFACB1D9FB315582B20D1B41476FC43AC4680647259
FE9B51371223446C82E0BBAEA132913E2B96EA11950C450F25854EE4FA4921193C8F1C66D61B826
5C7072B046F0C532141D51D9919C80733C1BD3C5A6D77CCB3A1938C95C1E4E866D1D65C78297B
3B32CA3C4143E6A215C609A36A1B13BAA17981D42B7FF4C715AC806DC491560032A5A2BB30E476
A266C6E4A4A065D9698DB08132608136082689B1B648B49063C98324706A43876507FC690893900F
8166E1D52ADCC44848D864B8BA0933B32DAA63435F11065915C5D5D879EDCC136FA8515B0260B9
536C316120B4904921805521C0232DA126E2A9C5323976ABC73B5AB892E59B01A194B6446C1B732
17FBBA855AC887AFA58A8F15A768EF968D775267E050150C7A8237C1024F9421C210D97907B2A14
4736E3B58E01C48947B1C62655E380256491CEBE400F52493D9033A29D6C9AE80B33E8C38584B30
A31A37B15E85F5E82B73157408A5399A957190CBA905F1713C9ACA53AA32923CACB8269436A56B
C02932189C7139D8463D0DC540621875E9A7DFEF020E7E83696C7612BC2A7DE7148D7075C31F25
7766301FA5A06E00582DCE15C6FB3195AF43078D79B110B5DC0789BE3A32132737FEB247DF8D22
16F272C3D5BA700C52ED7E3A2795CCFE3072CD1D1B533939AC58A065A8A9FE85441ED291D4386
9E25307737A5155A095C5056A204326F944A9FC8F7035C931A61033CC2DB6ECEC08AAE0045A654
2080C1A7BBD699902A2ECBE3A1F9AA95FC6222C6C6AB7BF96EDB3450E029B0CE104DBBFA0A49
B8044EE1BB63B33BCFAA8A6A450194CC577D490A549B6C42D6816F444AAE8B8A4F62AFFD17664
E957E2BA245A864BAB8D28E82CAB9FAC3BCD707AAC19A5EE58A3925DC7059BA1B9E7A798EC98
7E1924342C54E5D935A6699B6D9AA78E4738B0CEC6C57D8228C86881A14A3B57065BBE5653BD92
B5BD5A9888B1B1E9B017ECA1A788CBC3A283A9F7B00D3F55B138CA26C33CD2CA3BA632A073559
6162AA6C9D10968F53305D35CEC4395CA0489FF631A3EF661264A22ACB535DCC1313CD6917BFE
3B5B8FC3B8016CA0DFB151623C92F95701DCC4140459B52EB796FDA60FB429651052D2C16B550F
A035CC1374C87A439077713769C728B766BC75A609B70BBC14AF84A3C70519C3211ACAD58BF7C1
4C15F1AA86FCC1DF55580A2F9BBBC31B6B0EAAB379481CE9966F7FA6487C611BA5B3E8F8307FA
35CD5248C8AB351B63ABC4BA005871A97751F01E5143302C757A65E411AF7B26F22076C7A1CF7E
2B0055E80D7116CA2B3056BF8754BCFA9095D0CE99715CC4512F10125C5A500DABB7C11F5B0408
377900621BE851B7A3602576650B84191749F30AC635A9E2212400769DE8E1915F26BA198AF8E53D
ACB598711738DA8C583A388E027A59

dk =
E6F634F0A3771ED789D6842321E147B6010ADA7B6B0B105949F90AEBECCA062C494743AD3BE35
A23C9BAA0E4CB83E65C9EB8BBC42433F0AC671ED9544A37C0CD704D1C24343ACA7BEBB538C70
51186BC624A9A98F536874C3A15E145732A591ECE600C78028426BB8A7AF22757296EB968B5E355
3721DB6BD507A5BE5742F45AA3E134383499400BDAC0C8607FAD647F41A2B547A631F971224C93
570141C844662F96D9A3A32B6DB1C93F2A2B82E6B370B7D91EB9F9B8D551C8C2F3CEA6C2711379
969467ABF7041848BB05E619A7B63CC2F9DAC97C05457911470EC7B4CD380528D695604B6F04239
4524C5C77415C3CA15FA64A9E16B240ED12CFF0A48AF72743916C469593B89D8510BFEC58A0532
948D80D32A11FCE265ACBE6C21751507FD72366EB5EDD650344268757A715F27B28BEB4CF80347
DCFCA34BB92A046955AA39590C2397A223ACB42B94B938329CB6A9C4253137531C54D2100966B9
A8C96110A412AD2DC4480FAB04D05B93087AE8E4A1616526A48F38CAA920C6F495CA9E48877655
1630010C46C375DE52AA5EC23CCE9BCCE6663F1678B9F4325D1A961D551A02B046320F52272382
```

KEYGEN DONE

encap tcld = 26

ct =
4EE24D9E0858B36DC755A9389F4FDBF438DB8FBFDDD2E2A41FBFE7313693E87B2BD86
A2A5C95286840A2E477F4AAC12F28319D892C30FE9A120A09713369A17D5EC459C7E5DC
D402F9049BF6FF0F7D07A7F18D4C1E3E0429BF6D501EEDD33E114423A5C469273809610
1EC79233F20C58A6EA7E855C4E608DA7C9EE086EEAFE296F3214BD0B9264AE18069342
AB493BBEE267401FEDAE19D5F9224A11D911505CB3200C65F17C91F88FBE25621402C93
9F071C48D6BC6ACD207C8C215E2FF23AB5FAD94F2CE61003C99DE15195B36ADE088640
43CAEA49D4EA1D550978C49455B06AFB4BA4F0C178D30F953D47F0258370FE8E6868020
85FDCB25598DE439CEEA186F848AE2E4ACF526E03755D0A89A941A08C5610C96DFB3F4
7C693E7E899E1DD38DFD9126F3D05751234ABB82D2BBA4D347823B97293AE2B58FD3C7
1D2216797DA1CA453D600C5A9A6B36CA23B9450894A894A42C40B38871F1E9E84B40A04
112CC9A4B6E7B8DE59CAE5D8389687DBB1EA078F3CC93DC5D0788A671530D9FFDE5F2
CB6A558AB38E038D59ED1EF1FEE381AFB6DE0A77E1333FC5B0FE3EA2DCDB84FB33795
449AC6DC322F4E5E09B2A462D3750BA426C04271903CFE9A3F02221B69E474F24B7DB60
8492C01AAC3F2FA472A2469B6A8A949A67C567D139D6F40B00DF1AAC1E6456FFAA646E0
DEBD79907905F28076F78ECB3DD1A6493A40F3D1F8D89C71AD15D22162264D0AA73581
AF7A77EEA43B856370182348034810E60A902604B724E09EA4638AAB039982A64EF02C9C
21259A381BFD3BF298EFE4285CB538F79BE01CCEE1A4A903A80132942AC6BCD4425BD1
97EF014E7DBEFFDFDD5AAD00DA2A5E8DE4FFF1E8E97F23F3F4BE38970AC09CB99D252
CF2736059CFDE177EA6B36EA329B9DC541C3DCE61FAC504908857184C936C1A1C4AC0
C8DDE9A66B4F3AAADEAB2DBEB2F2E6422A49A6B9C223DF25A34720717253DA6DCFC0A
825553A1D2884E01A97060886FC30A1AD50E52C46F6F9B87274AD6369FCE4FDF3FF6A5D
62D4AC82CBCECE902E4DE4214B71A2CA547C536380DCA26F63F2BE71B01EAC68F42AD
1399D390404AA5539A5995254836FE6CF59C490B13CAC01C69E50CDC0EBA1C2F9978F42
23AA38DA33639ABE933B40F6E2751515EFAC9896465041D3922A4B043897083F7CC6DAF
1FD87B970BB6D1D5E48B599FE6CD2AFBA5FCE17E4BD4B0217E879FAE34FC227EC132D
6F7141A48D14DED57729164A1D2B0BEF89A16244EE114AB4E108648B7E80A7DD502B0D
BD0C38AF1D85AF6966BFB8E54237000A3F18C48EFEB93ED62B41A9B341361D0D7AE75D
63A3A34DCEE329996AB0630553F38BC0DA148419162A31D386745C818E2691411A7927E4
9822AAE9EA918BF49D6809290B57C1BA5EC212F135AC0B8EE945C0510CFED4DB5A84B6
17BA1525E996486728E1DB87B6E99CC9A7EC4399B063D3943FE4EB1933E90365A4916CA
8D67D064DF7D5B8DFC51E337247CAC3CFFAEDC5523276CDCB82B130D8E1C3FB6D7D6
9

ss =
B2425299020BCF563B8EBE0512F0479941335A75A32B8D10BFF60E5548B64672

ENCAP DONE